

MSMS 408 : Practical 03

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Exam Roll No. : 24419STC053

June 5, 2026

➔ Question 1

In a competitive examination, a student attempts 20 independent multiple-choice questions. Each question has four options, only one of which is correct, and the student answers each question randomly. The probability of answering any question correctly is therefore $1/4$. Let X denote the number of correct answers of the student.

- (i) Assuming independence of questions, model X using an appropriate probability distribution.
- (ii) Write down its probability mass function, compute the probability that the student answers exactly 6 questions correctly.
- (iii) Find the expected number of correct answers.
- (iv) Simulate observations of X using the inverse transform method to compare empirical and theoretical probabilities.

➔ Theory

- 1** In the given set-up, X is said to have a Binomial distribution with parameters $n = 20$ and $p = \text{probability of success} = \frac{1}{4}$.

We write

$$X \sim \text{Binomial}\left(20, \frac{1}{4}\right).$$

- 2** The Probability Mass Function of X is given by

$$P[X = x] = \binom{20}{x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{20-x}; \quad x = 0, 1, 2, \dots, 20.$$

Thus, the probability that the student answers exactly 6 questions correctly will be

$$\begin{aligned} P[X = 6] &= \binom{20}{6} \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^{20-6} \\ &= \binom{20}{6} \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^{14} \\ &\approx 0.182. \end{aligned}$$

- 3** If $X \sim \text{Binomial}\left(20, \frac{1}{4}\right)$, $E[X] = 20 \cdot \frac{1}{4} = 5$. So expected number of correct answers of the student is 5.

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The inverse transform method to generate samples from a discrete random variable X is as follows.

- I. Generate $u_i \sim U(0, 1) \forall i = 1(1)n$.
- II. Map $x_i = k$ whenever $F(k-1) < u_i \leq F(k) \forall i = 1(1)n$ where $F(\cdot)$ is the CDF of X .

We are actually doing

$$X = \min\{k : F(k) \geq U\}.$$

➔ R Program

```
set.seed(22)
```

```
n <- 10000
```

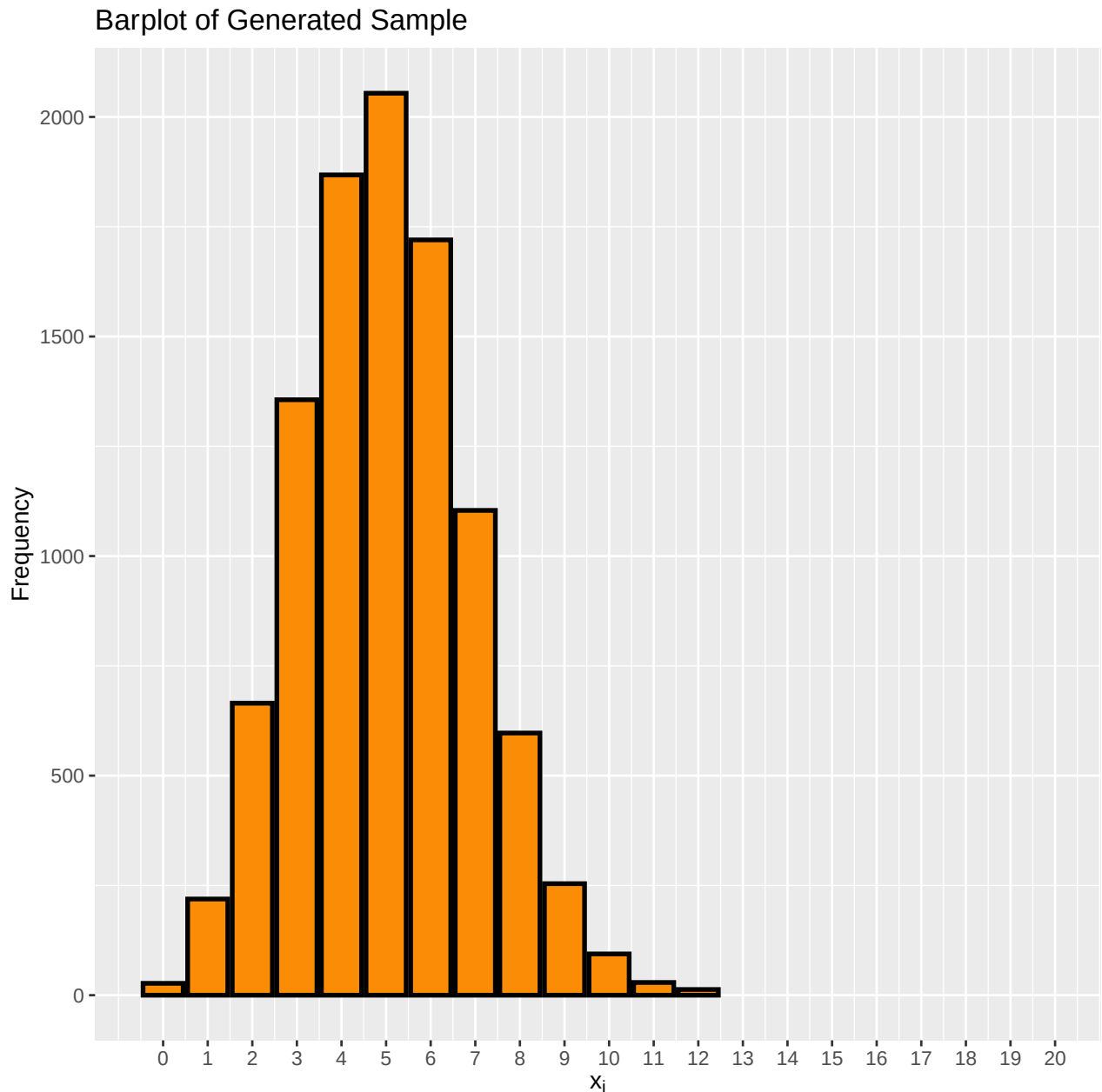
```
u <- runif(n)
```

```
CDF_F <- pbinom(q = 0:20, size = 20, prob = 1/4)
```

```
x <- findInterval(u, CDF_F, left.open = TRUE)
```

Let us have a barplot of the generated sample.

```
data.frame(x) %>%  
  ggplot(aes(x = x)) +  
  geom_bar(fill = "#FB8C06", color = "black", linewidth = 1) +  
  scale_x_continuous(limits = c(-0.5,20), breaks = 0:20) +  
  labs(x = expression(x[i]), y = "Frequency",  
       title = "Barplot of Generated Sample")
```



Most of the sample values are in between 1 and 13. Values in the tail have a very small probability mass, so with $n = 10000$ only, it is unlikely to observe them and exactly that has happened to be the case.

The empirical probabilities are obtained as

$$\hat{P}(X = x) = \frac{\text{count}(x)}{n}; \forall x = 0(1)20.$$

```
emp_prob <- table(factor(x, levels = 0:20)) / n
emp_prob <- as.numeric(emp_prob)
```

```
theo_prob <- dbinom(x = 0:20, size = 20, prob = 1/4)
```

```
df <- data.frame(x = 0:20,
  empirical.probability = emp_prob,
  theoretical.probability = theo_prob)
```

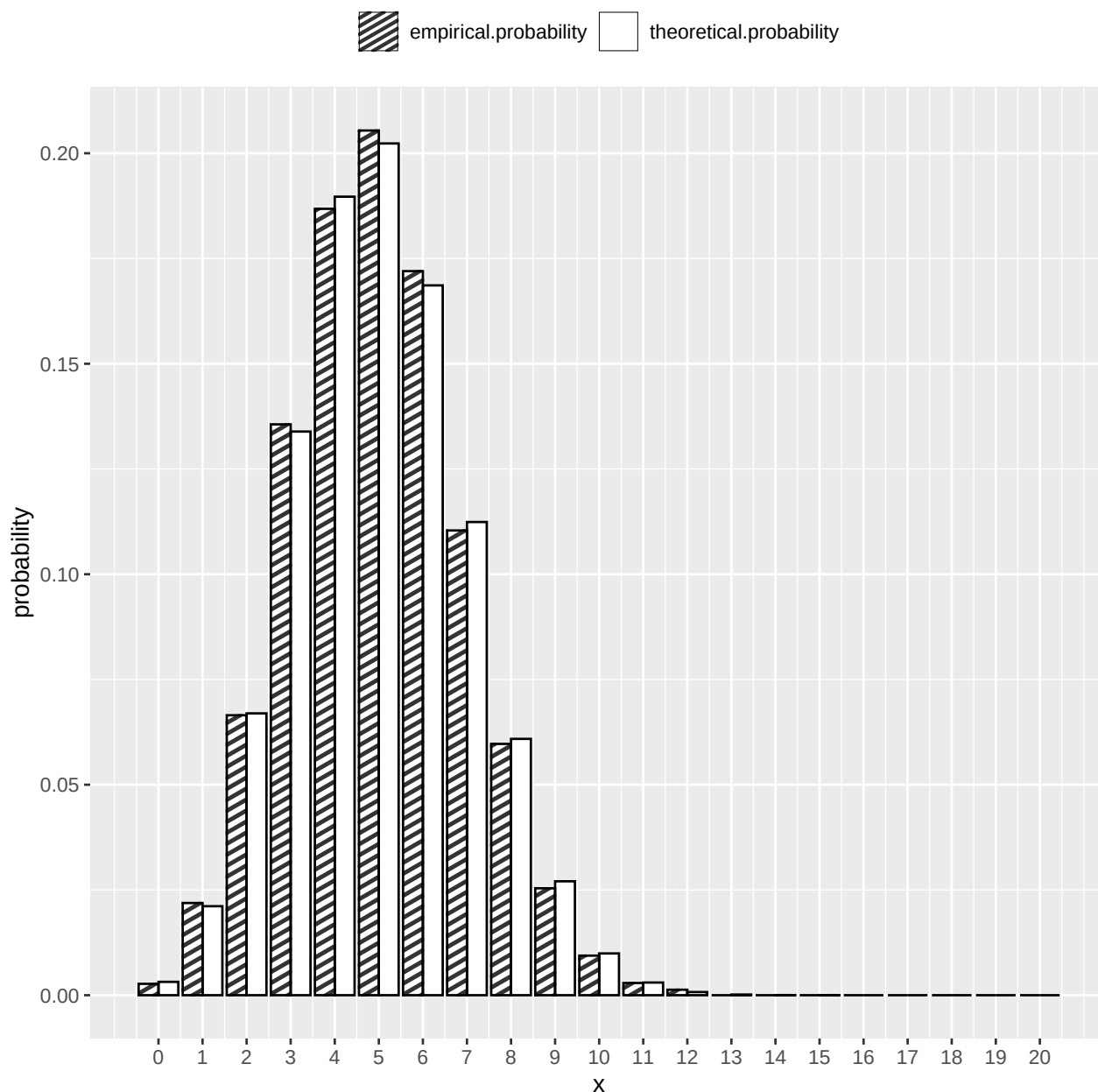
Table 1:

x	empirical.probability	theoretical.probability
0	0.00270	0.00317
1	0.02190	0.02114
2	0.06650	0.06695
3	0.13560	0.13390
4	0.18680	0.18969
5	0.20540	0.20233
6	0.17200	0.16861
7	0.11040	0.11241
8	0.05970	0.06089
9	0.02540	0.02706
10	0.00940	0.00992
11	0.00290	0.00301
12	0.00130	0.00075
13	0	0.00015
14	0	0.00003
15	0	0.000003
16	0	0.0000004
17	0	0.0000000
18	0	0
19	0	0
20	0	0

```
df_melted <- df %>%
  pivot_longer(cols = c("empirical.probability", "theoretical.probability"),
    names_to = "probability_type",
    values_to = "probability")
```

Let us have a grouped barplot from comparison of empirical and theoretical probabilities.

```
df_melted %>%  
  ggplot(aes(x = x, y = probability, pattern = probability_type)) +  
  geom_col_pattern(position = "dodge",  
                  pattern_spacing = 0.01,  
                  fill = "white", color = "black") +  
  scale_pattern_manual(values = c("stripe", "none")) +  
  scale_x_continuous(limits = c(-0.5, 20.5), breaks = 0:20) +  
  theme(legend.position = "top",  
        legend.title = element_blank())
```



➔ Conclusion

Empirical probabilities very much agree with theoretical probabilities.

➔ Question 2

A fair die is rolled 100 times and let X denote the number appearing on the top face in each roll. Generate the 100 outcomes using a suitable random number method, compute the frequency of each face (1–6), and compare the observed relative frequencies with the theoretical probability $1/6$ for a fair die.

➔ Theory

In the said set-up, X is said to have a Discrete Uniform distribution with support $\{1, 2, 3, 4, 5, 6\}$. The Probability Mass Function of X is given by

$$P[X = x] = \frac{1}{6}; \quad x = 1(1)6.$$

We shall use Inverse Transform method to generate sample from X . Recall that, the CDF of X at $\{1, 2, 3, 4, 5, 6\}$ is

$$F(x) = \frac{x}{6}; \quad x = 1(1)6.$$

➔ R Program

```
n <- 100; u <- runif(n); CDF_F <- (1:6) / 6
```

```
x <- findInterval(u, CDF_F, left.open = TRUE) + 1
```

```
df1 <- data.frame(x = 1:6, count = as.numeric(table(factor(x, levels = 1:6))))
```

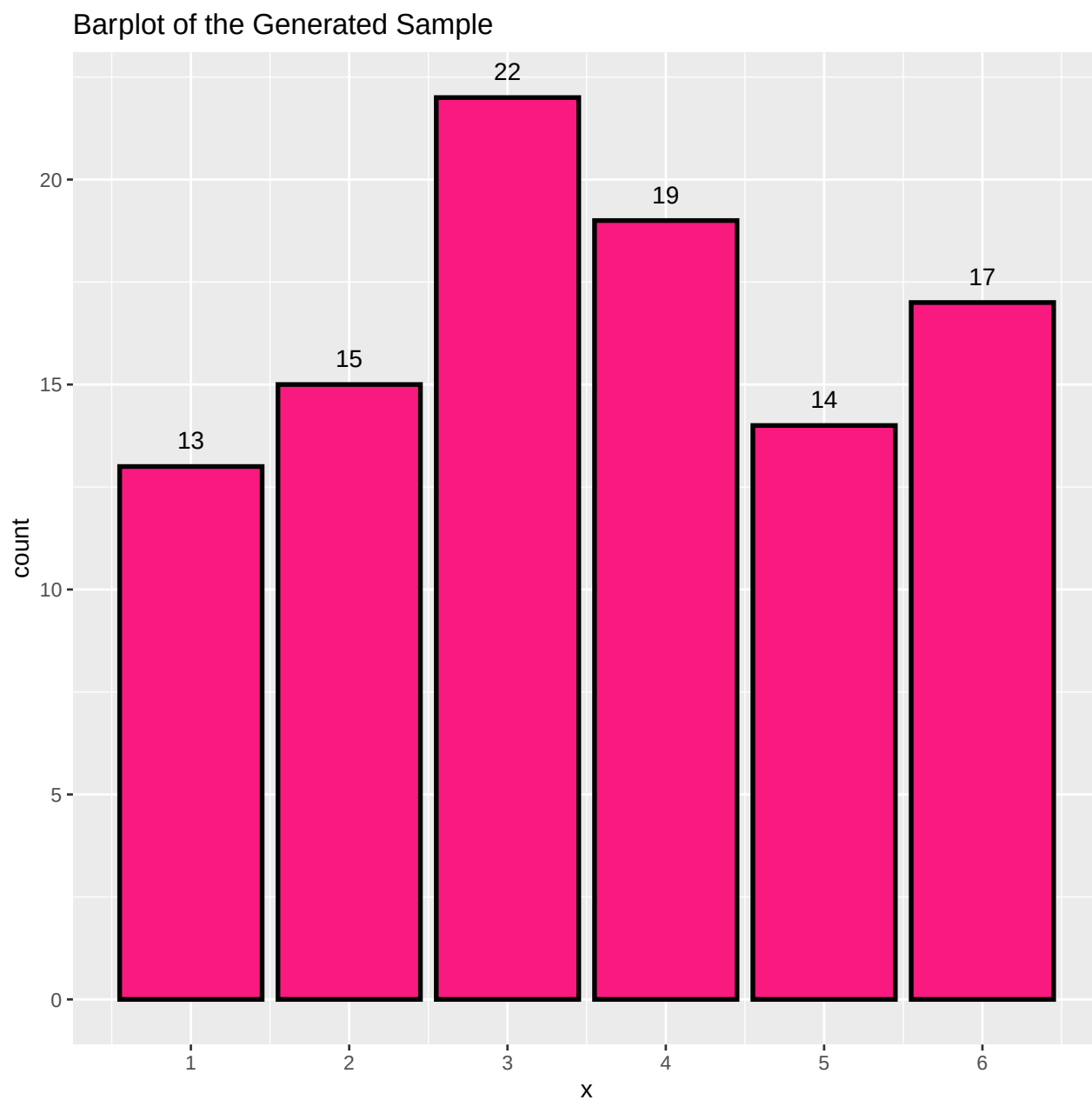
```
stargazer(df1, summary = FALSE, rownames = FALSE, label = "Table 2")
```

Table 2:

x	count
1	13
2	15
3	22
4	19
5	14
6	17

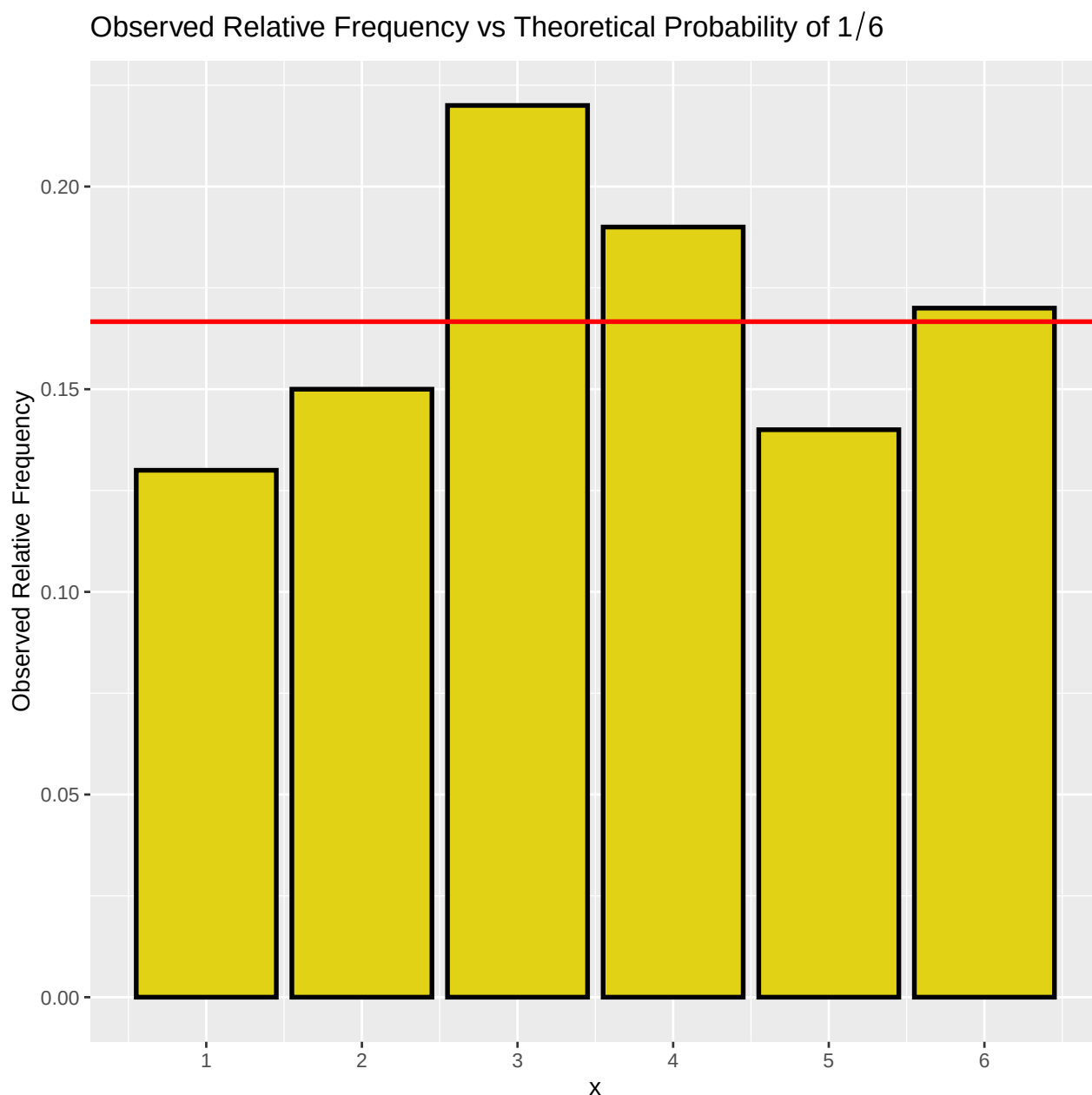
Let us have a barplot of the generated sample.

```
df1 %>%  
  ggplot(aes(x = x, y = count)) +  
  geom_col(fill = "#F81A7F", col = "black", linewidth = 1) +  
  scale_x_continuous(breaks = 1:6) +  
  geom_text(aes(label = count), vjust = -1, size = 4) +  
  labs(title = "Barplot of the Generated Sample")
```



Here we have a plot to compare the observed relative frequencies and theoretical probability of $\frac{1}{6}$.

```
df1 %>%  
  ggplot(aes(x = x, y = count / n)) +  
  geom_col(fill = "#E1D216", col = "black", linewidth = 1) +  
  scale_x_continuous(breaks = 1:6) +  
  geom_hline(yintercept = 1/6, col = "red", linewidth = 1) +  
  labs(y = "Observed Relative Frequency",  
       title = expression("Observed Relative Frequency vs Theoretical Probability of" ~ 1/6))
```



➔ Conclusion

 Observed relative frequencies are consistent with theoretical probability.